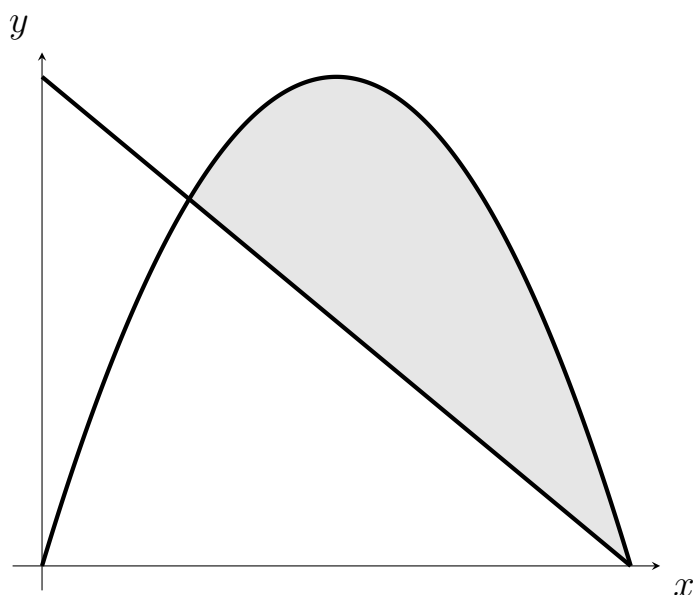
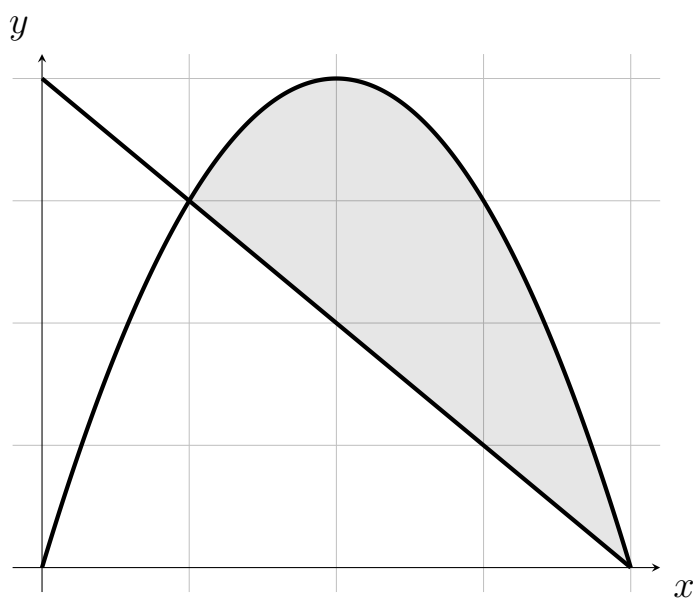


6.1 Area Between Curves

Suppose we wish to find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.



How can we approximate this area?



Area Between Curves

Let R denote the region between $y = f(x)$ and $y = g(x)$ between the vertical lines $x = a$ and $x = b$ given that $f(x) \geq g(x)$ on $[a, b]$.

The area of R is defined by

$$A =$$

where we divide R into _____ strips of equal width, _____
and height _____.

What is x_i^* ?

Area Between Curves

The area, A , of the region bounded by the curves $y = f(x)$ and $y = g(x)$, and the vertical lines $x = a$, $x = b$, where

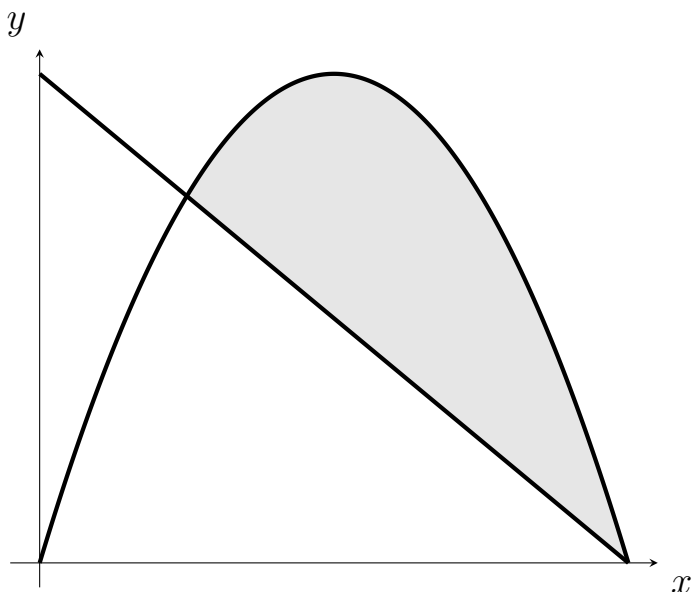
- f and g are _____ for all x in $[a, b]$ and
- $f(x) \geq g(x)$

is given by

$$A =$$

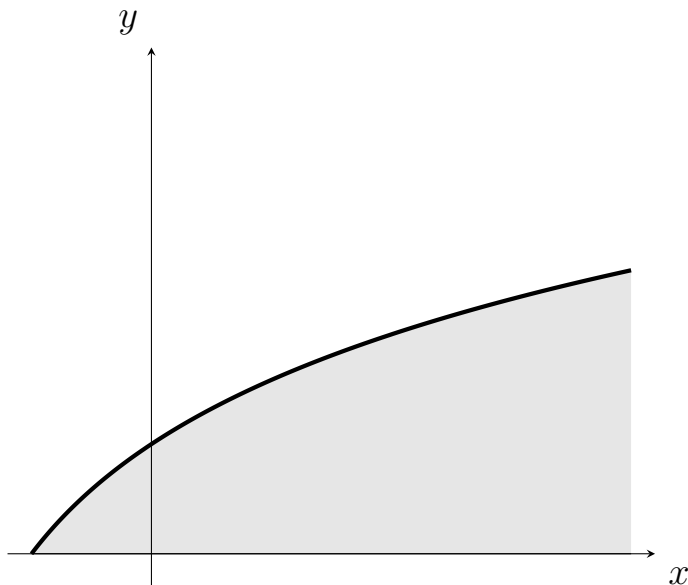
Example 1

Set up an integral to find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.



Example 2

Set up an integral to find the area under the curve $y = \ln(x + 2)$ on the interval $[-1, 4]$.



Why is Example 2 an area between curves problem?

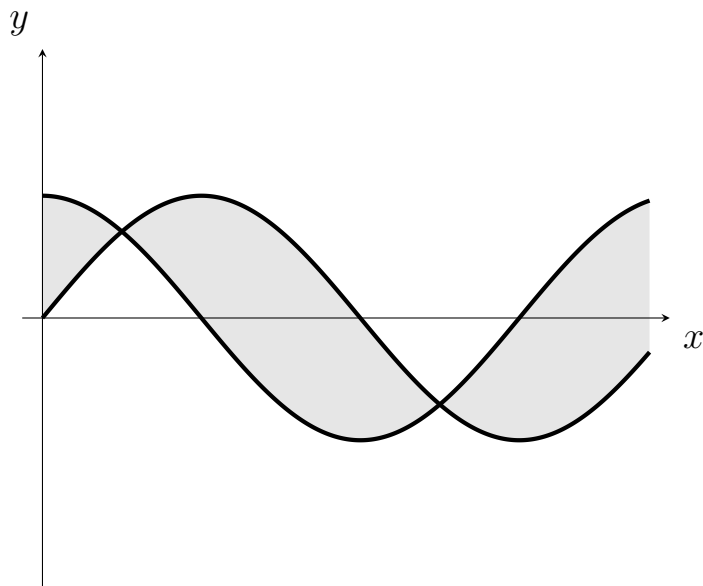
What happens if the two functions between which we wish to find the area, intersect on the interval $[a, b]$? Why would this pose a problem?

Finding the Area Between Two Curves

1. Identify the domain.
2. Find where the curves intersect.
3. Determine the subintervals over which you need to calculate area.
4. Determine whether $f(x) \leq g(x)$ or $g(x) \leq f(x)$ on each subinterval.
5. Set up the integrals accordingly, and evaluate.

Example 3

Set up an integral to find the area enclosed between $f(x) = \cos(x)$ and $g(x) = \sin(x)$ on $[0, 2\pi]$.



Can we find the area between two functions if we are not given a fixed domain?

All of the examples that we have seen so far have been functions of x . Suppose we wish to consider functions of y instead. Why might this be necessary?

Example 4

Set up an integral to find the area enclosed by $x = y^4$ and $y^2 = 2 - x$.

Extra Example

Try these problems on your own and compare your answers to those from the videos.

1. Set up an integral to find the area of the triangular region enclosed by the points $(0, 0)$, $(2, 2)$, $(5, 0)$ with respect to both x and y .

2. Find the area enclosed by the curves $4x^2 + y = 4$ and $x^4 - y = 1$.

3. Set up an integral to find the area of the region enclosed by $y = 2\sin(x)$ and $y = \sin(2x)$ on $[0, 2\pi]$.